

Mixture of Flows

Notes on Multi-Teacher Diffusion Distillation
via Learnable (Teacher×Source×Time) Routing

Flow Factory

Abstract

We study the problem of distilling K specialized diffusion teachers $\{v^{(k)}\}_{k=1}^K$ (each fine-tuned for a different reward objective) into a single student that achieves *teacher-level in-domain performance on every reward* while picking up cross-objective gains on out-of-distribution (OOD) rewards. We make three observations:

(i) Standard On-Policy Distillation (OPD) for diffusion decomposes into a path-wise MSE term and a policy-gradient (PG) term; under the standard OPD heuristic (the PG term shares the pathwise fixed point and empirically contributes mostly variance), *plain velocity MSE suffices*.

(ii) Naive multi-teacher OPD has two natural failure modes: *route-by-source* caps the student at single-teacher performance with no OOD gain, and *averaging* loses in-domain by Jensen’s inequality with only a small OOD bonus.

(iii) We can construct a *potentially better target* by taking a learnable convex combination $v_\lambda = \sum_k \lambda_{k,t,s,c} v^{(k)}$ with weights conditioned on (teacher, time, source, prompt). Under the shared-marginal approximation (teachers are LoRA fine-tunes of one base model), any *affine* combination of valid flow-matching velocities is itself a valid flow-matching velocity —convexity is an optimization choice, not a validity requirement (cf. CFG) —so this v_λ is a legitimate diffusion model and can be tuned *by RL* (DiffusionNFT or Flow-GRPO) on the small mixing parameter space. The student is then distilled from v_{λ^*} via plain pathwise MSE.

The framework provides a *constructive* Level-1 in-domain guarantee: the student’s expected reward on every in-domain dataset is at least that of the corresponding single teacher. We also establish that strict OOD improvement is *not* guaranteed in general; a precise negative theorem with three formal obstructions is given, alongside three sufficient conditions (temporal complementarity, non-conflicting reward gradients, non-empty Pareto-improving region) under which OOD gains become provable. We close with the experimental program (three teachers trained to GenEval=0.958, OCR=0.953, PickScore≈24.13) and a list of failure modes observed during preliminary runs.

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1 Introduction

1.1 Motivation

Modern text-to-image diffusion fine-tuning has become a multi-objective problem: human preference (*e.g.* PickScore), compositional accuracy (*e.g.* GenEval), and text-rendering fidelity (OCR) are all desirable, each with its own reward model and training set. RL fine-tuning (Flow-GRPO, DiffusionDPO, DiffusionNFT) typically optimizes a single reward, so the practical pipeline is to train K *specialist* LoRA teachers $\{v^{(k)}\}_{k=1}^K$ separately—producing strong in-domain performance per reward, but no single deployable artifact that excels at all of them.

The natural follow-up question is: *can we distill these K teachers into a single student*, with the student matching each teacher’s in-domain strength while picking up cross-objective gains? We refer to in-domain performance as the score of reward R_s on prompt set $\mathcal{P}_s$ (the set the teacher $v^{(s)}$ was trained on), and OOD performance as R_j on $\mathcal{P}_s$ for $j \neq s$.

1.2 Failure modes of prior approaches

Two natural baselines, both descended from On-Policy Distillation (OPD), fail in symmetric ways:

- **Route-by-source.** Each prompt is routed to its source-matched teacher and the student fits only that teacher. The student’s performance is upper-bounded by the corresponding single teacher; *no* OOD gain because cross-teacher information never enters the gradient.
- **Averaging.** The student fits the arithmetic mean $\bar{v} = \frac{1}{K} \sum_k v^{(k)}$. By Jensen’s inequality (and the α -relation between velocity-MSE and KL distillation), the student’s in-domain performance *strictly degrades* below the single-teacher baseline; only marginal OOD bonus is observed.

Both modes leak from the same fundamental constraint: the student’s velocity field is rigidly fixed at every (x, t) once the target is specified. There is no slack to spend on *cross-objective synergy*, where one teacher might be best at R_s in early timesteps (layout) and another teacher best at R_j in late timesteps (texture).

1.3 Our approach

We build a *better teacher* before distilling. Specifically, let

$$v_\lambda(x, t, c) = \sum_{k=1}^K \lambda_k(t, s(c), c) v^{(k)}(x, t, c), \quad \sum_k \lambda_k = 1, \quad \lambda_k \geq 0, \quad (1)$$

with weights λ depending on the teacher index k , the timestep t , the source label $s(c)$, and possibly the prompt embedding c . We optimize the λ ’s (and only those) by RL on a composite reward *in the velocity-output space*, then distill the final v_{λ^*} into a single student LoRA via plain pathwise MSE.

1.4 Contributions and roadmap

1. **Methodological** (§3): A three-stage pipeline—OPD pathwise sufficiency $\rightarrow$ MoF target construction by RL $\rightarrow$ on-policy distillation—decoupling “find a good convex weighting” from “encode it in a single LoRA”. The mixing search space is just $K \times T \times S \sim 10^2$ parameters (LUT) or $\sim 10^6$ (router), tiny compared to the 10^7 – 10^9 student parameters.

2. **Theoretical** (§4): *Constructive* Level-1 in-domain guarantee (§4.1); *conditional* Level-2 in-domain strict-improvement under temporal complementarity (§4.2); a *negative* theorem on OOD strict improvement (§4.3) showing the asymmetry between in-domain and OOD, accompanied by three sufficient conditions; an analysis of NFT’s β -cancellation property (§4.4); and a discussion of distillation-gap erosion (§4.5).
3. **Experimental** (§5): A protocol with three Flow-DPPO teachers (GenEval/OCR/PickScore at 0.958/0.953/24.128 respectively), four mixing-module variants (LUT, LUT-SA, Time Router, adaLN/MLP Router), two RL options (DiffusionNFT, Flow-GRPO), and ablations on the OOD-bonus γ and initialization. We report a preliminary failure mode—the *PickScore-dominance* effect—in which neural routers asymmetrically inflate PickScore signal because PickScore’s `applicable_sources` covers all sources while the others are domain-specific.

2 Background

2.1 Flow matching and notation

Let $x_0 \sim p_{\text{data}}$, $x_1 \sim \mathcal{N}(0, I)$. Flow matching parameterizes a velocity field $v_\theta(x, t, c)$ such that the ODE $\dot{x}_t = v_\theta(x_t, t, c)$ transports the noise distribution to the data distribution. The standard training loss is

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{x_0, x_1, t} \left[\|v_\theta(x_t, t, c) - (x_1 - x_0)\|_2^2 \right], \quad x_t = (1 - t)x_0 + tx_1. \quad (2)$$

A teacher is a frozen $v^{(k)}$ (typically a LoRA snapshot of a base flow model). We index teachers by $k \in \{1, \dots, K\}$, sources by $s \in \{1, \dots, S\}$ (where each source has an in-domain reward R_s), timesteps by $i \in \{0, \dots, T - 1\}$, and per-source prompt sets by $\mathcal{P}_s$. We use $f_j^{(s)}(\lambda) \triangleq \mathbb{E}_{p \sim \mathcal{P}_s} [R_j(\text{ODE}(v_\lambda; x_T, p))]$ to denote the expected reward R_j achieved by mixing λ on prompt set $\mathcal{P}_s$.

2.2 OPD: pathwise vs. policy-gradient

Suppose the goal is to distill a teacher v^{tgt} into a student v_θ . Let q_θ be the student’s induced trajectory distribution and q_{tgt} the teacher’s. Minimizing the path-wise KL $\text{KL}(q_\theta \| q_{\text{tgt}})$ has the well-known decomposition (cf. standard OPD literature):

$$\nabla_\theta \text{KL}(q_\theta \| q_{\text{tgt}}) = \underbrace{\mathbb{E}_{q_\theta} \left[\sum_i \frac{1}{2\sigma_i^2} \nabla_\theta \|\mu_\theta^i - \mu_{\text{tgt}}^i\|^2 \right]}_{\text{pathwise}} + \underbrace{\mathbb{E}_{q_\theta} \left[\sum_i (\log q_\theta - \log q_{\text{tgt}}) \nabla_\theta \log q_\theta \right]}_{\text{policy-gradient (REINFORCE)}}. \quad (3)$$

Remark 1 (Heuristic: PG term treated as variance-only). We adopt the standard OPD working assumption that the policy-gradient term in (3) contributes *variance but no useful direction* once the pathwise term is applied, and may therefore be dropped. We stress that this is a *heuristic*, not a theorem. The score-function identity gives $\mathbb{E}[g_i] = 0$ for $g_i := \nabla_\theta \log q_\theta$, so the PG term $\mathbb{E}[r_i g_i]$ (with $r_i := \log q_\theta - \log q_{\text{tgt}}$) equals the *covariance* $\text{Cov}(r_i, g_i)$ —which vanishes only if r_i and g_i are uncorrelated, an assumption that is plausible near convergence (where $r_i \rightarrow \text{const}$) but not justified in general; indeed the PG term is by construction exactly the gap between the pathwise gradient and the full KL gradient. The honest statements are: (a) at the fixed point $\mu_\theta = \mu_{\text{tgt}}$ the PG term vanishes, so both losses share their optimum; (b) empirically, OPD-style training with the pathwise term alone reaches the same quality as the full gradient while being far lower variance. Our pipeline relies on (a)+(b), and §6 includes a `pathwise_only` vs. `both` ablation to verify (b) in our setting.

Practical consequence: distillation in our pipeline reduces to plain velocity MSE (no log-prob, no importance ratio, no clipping). This is what `trainers/mof/distill.py` implements (see `mof_two_stage_pipeline.tex`).

2.3 Affine combinations of flow matching velocities under shared marginals

Proposition 1 (Shared-marginal affine closure of FM). Let $v^{(1)}, \dots, v^{(K)}$ be velocity fields that transport the *same* marginal path p_t , i.e. each satisfies the continuity equation $\partial_t p_t + \nabla \cdot (p_t v^{(k)}) = 0$ with a common p_t . Then for any *affine* combination, $\lambda \in \mathbb{R}^K$ with $\sum_k \lambda_k = 1$ (negative weights and weights > 1 allowed),

$$v_\lambda = \sum_k \lambda_k v^{(k)} \quad (4)$$

also satisfies the continuity equation for p_t , hence is a valid flow-matching velocity field for the same marginal path.

Proof. For fixed p_t the continuity equation is linear in v : $\partial_t p_t + \nabla \cdot (p_t \sum_k \lambda_k v^{(k)}) = \sum_k \lambda_k [\partial_t p_t + \nabla \cdot (p_t v^{(k)})] + (1 - \sum_k \lambda_k) \partial_t p_t = (1 - \sum_k \lambda_k) \partial_t p_t$, which vanishes iff $\sum_k \lambda_k = 1$ (for non-stationary p_t). $\square$

Remark 2 (What the affine constraint does and does not require). Two consequences deserve emphasis. **(i) Convexity is not required.** Validity needs only the *sum-to-one* (affine) constraint, not $\lambda_k \geq 0$. Classifier-free guidance is the canonical example: $v_{\text{CFG}} = (1 - w) v_{\text{uncond}} + w v_{\text{cond}}$ with $w > 1$ is an affine, non-convex combination that works extremely well in practice. Restricting MoF to the simplex Δ^{K-1} (softmax) is therefore an *optimization choice*, not a validity requirement —this motivates the weight-normalization ablation (softmax / affine / unnormalized) in the implementation. Conversely, dropping the sum-to-one constraint entirely *does* break validity: if $\sum_k \lambda_k = c \neq 1$ the residual term $(1 - c) \partial_t p_t$ remains, i.e. the velocity field is rescaled —an effective time-reparameterization that pushes samples off the marginal path. The unnormalized MoF mode therefore pairs raw weights with a soft penalty anchoring $\sum_k \lambda_k \approx 1$. **(ii) The shared-marginal hypothesis is an approximation.** Our teachers are LoRA fine-tunes of one base model, so their marginals are close but *not* identical —RL fine-tuning shifts marginals by design. Without shared marginals, the plain mixture $\sum_k \lambda_k v^{(k)}$ does *not* transport the mixture of marginals $\sum_k \lambda_k p_t^{(k)}$: the continuity equation is linear in the pair (p, pv) , so the mixture marginal is transported by the *density-weighted* velocity

$$\tilde{v}_\lambda(x, t) = \frac{\sum_k \lambda_k p_t^{(k)}(x) v^{(k)}(x, t)}{\sum_k \lambda_k p_t^{(k)}(x)}, \quad (5)$$

which differs from v_λ wherever the $p_t^{(k)}$ disagree. For MoF this is benign in practice: v_λ is treated directly as a sampler (its ODE is well-posed regardless) and is optimized by RL on the rewards it actually achieves, so no claim about transporting a particular marginal is needed. But “ v_λ is a legitimate diffusion model” should be read as exact under shared marginals and approximate (first order in the inter-teacher marginal gap) otherwise.

This is the key structural fact that enables MoF: *any* per-timestep mixing v_λ defines a well-posed sampling ODE whose trajectories can be sampled, scored, and distilled —exactly valid as an FM field under shared marginals, approximately so for LoRA-finetuned teacher ensembles.

2.4 Failure modes of multi-teacher OPD

Let $f_j^{(s)}(\theta)$ denote the student’s expected reward R_j on prompt set $\mathcal{P}_s$ after distillation.

Route-by-source. Train K separate students or a single student that switches teacher by $s(p)$. The optimal student velocity is $v_\theta^*(x, t | p \in \mathcal{P}_s) = v^{(s)}(x, t)$, hence $f_s^{(s)}(\theta^*) \approx f_s^{(s)}(v^{(s)})$ (in-domain matches teacher exactly, modulo distillation gap), and $f_j^{(s)}(\theta^*) \approx f_j^{(s)}(v^{(s)})$ (OOD just inherits teacher s ’s OOD score—no cross-teacher transfer). *No OOD gain.*

Averaging. Train a student to fit $\bar{v} = \frac{1}{K} \sum_k v^{(k)}$. The optimal student is $v_\theta^* = \bar{v}$. By Jensen’s inequality applied to sufficiently sharply-peaked rewards, $f_s^{(s)}(\bar{v}) < f_s^{(s)}(v^{(s)})$ for every in-domain pair (s, s) (*strict in-domain degradation*). OOD often improves slightly because $\bar{v}$ blends teacher behaviors, but the gain is small and not in general guaranteed.

Both failure modes share a structural cause

The student’s optimal velocity is rigidly determined by the target construction: route picks a vertex of the teacher convex hull ($v^{(s)}$), averaging picks the centroid. Neither uses the per-timestep slack that Proposition 1 grants. To exploit it, the weights must be *learned* based on what actually helps the reward landscape—this is the MoF idea.

3 Method: Mixture of Flows

3.1 The MoF construction

Let

$$v_\lambda(x, t, c) = \sum_{k=1}^K \lambda_k(t, s(c), c; \psi) v^{(k)}(x, t, c), \quad \lambda_k = \frac{\exp(\ell_k/\tau)}{\sum_{k'} \exp(\ell_{k'}/\tau)}. \quad (6)$$

Here ψ is a small set of *learnable mixing parameters* (logits ℓ shaped through softmax). The teachers are frozen and detached.

We instantiate λ in five flavors of increasing expressiveness, each realized in `trainers/mof/common.py` by a distinct `MoFMixingModule` subclass:

Module	Form of ℓ	Notes
LUT	$\ell \in \mathbb{R}^{K \times T \times S}$	Per-source & per-step lookup; $K \cdot T \cdot S \sim 10^2$ params.
LUT-SA	$\ell \in \mathbb{R}^{K \times T}$	Source-agnostic; broadcast over S . Diagnostic baseline.
Time Router	$\ell = \text{MLP}(t)$	Continuous in t ; no prompt conditioning.
adaLN Router	$\ell = \text{MLP}(\text{adaLN}(t, c))$	adaLN-fused (DiT-style).
MLP Router	$\ell = \text{MLP}([\text{TE}(t) \ c_{\text{pool}}])$	Concat-then-MLP.

We collectively denote the parameter set ψ (whether $K \times T \times S$ logits or router weights). The number of parameters in ψ is at least four orders of magnitude smaller than the number of student parameters.

3.2 Two-stage pipeline

Stage 1 (RL on ψ). We optimize ψ by RL on a composite reward, treating v_λ as the policy. Because v_λ is a valid FM under shared marginals (Prop. 1; approximate for LoRA teacher ensembles, Remark 2), any RL algorithm that supports a flow-matching policy applies. We use **Dif-fusionNFT** as the primary algorithm (path-wise reconstruction loss reweighted by advantage; details in §3.3.1 and `mof_two_stage_pipeline.tex`). Alternative: **Flow-GRPO** (PPO-style ratio with a stochastic CPS scheduler) is supported as an ablation.

Stage 2 (distill v_{λ^*} into a student LoRA). After Stage 1, ψ^* is frozen and defines the *router-induced teacher* $v^{\text{MoF}^*}(x, t, c) = \sum_k \lambda_k(t, s(c), c; \psi^*) v^{(k)}(x, t, c)$. We train a single student LoRA θ by on-policy MSE:

$$\mathcal{L}_{\text{dist}}(\theta) = \mathbb{E}_{(x_t, t, c) \sim q_\theta} \left[\|v^\theta(x_t, t, c) - v^{\text{MoF}^*}(x_t, t, c)\|_2^2 \right], \quad (7)$$

where q_θ is the student’s own trajectory distribution (Stage 2 is on-policy: the student rolls out, then at each visited (x_t, t) the teacher mixture is computed and matched). Because the PG term is dropped per the heuristic of Remark 1, this MSE is the entire distillation loss.

Why two stages? Stage 1 finds ψ^* in a tiny parameter space using small batches of RL signal; Stage 2 amortizes the K -teacher inference cost into a single LoRA forward. At deployment, the student has the same inference cost as any single teacher, while behaving as the optimized convex combination.

3.3 Algorithmic choices

3.3.1 NFT vs. Flow-GRPO

DiffusionNFT optimizes ψ via reward-weighted self-normalized MSE on x_0 reconstruction:

$$\mathcal{L}_{\text{NFT}} = \frac{r \cdot \mathcal{M}(v^+) + (1 - r) \cdot \mathcal{M}(v^-)}{\beta} \cdot A_{\max}, \quad \mathcal{M}(v) = \frac{\|x_t - \sigma v - x_0\|^2}{\text{sg}(\|x_t - \sigma v - x_0\|_1/d)}, \quad (8)$$

where $r = \text{clip}(\frac{A}{2A_{\max}} + \frac{1}{2}, 0, 1)$ is the normalized advantage, $v^+ = \beta v_{\text{new}} + (1 - \beta)v_{\text{old}}$ and $v^- = (1 + \beta)v_{\text{old}} - \beta v_{\text{new}}$ are positive/negative interpolated velocities, and v_{old} is the EMA-shadow mixing (off-policy). Detailed derivation: `mof_loss_gradient_analysis.tex`.

Flow-GRPO instead uses the PPO-clipped log-prob ratio $\rho = \exp(\log p_{\text{new}} - \log p_{\text{old}})$, requires a stochastic dynamics (CPS / Flow-SDE), and is on-policy in the strict sense.

Choice: We use NFT as the default. NFT *converges faster* empirically on the small-parameter MoF search (consistent with its self-normalized weighting; the gradient signal is large at the start and self-attenuates as $x_0^{\text{pred}} \rightarrow x_0$). NFT also doesn’t require log-prob computations, halving the memory footprint of Stage 1.

3.3.2 LUT vs. Router

LUT-style modules give *decoupled* per-source per-timestep weights $(\Lambda_{k,t,s})$, making the optimization landscape $T \cdot S$ independent simplex problems. Routers give continuous t -flexibility (decouple training and inference T) and prompt conditioning, but introduce a parameter-sharing bias whose pitfalls we discuss in §6.

4 Theoretical Analysis

We work in the per-set MoF setting: each $\mathcal{P}_s$ has its own ψ_s (or its slice $\Lambda_{\cdot, \cdot, s}$), so the optimization decouples across sources.

4.1 Level-1 in-domain guarantee (constructive)

Theorem 1 (In-domain $\geq$ single teacher, up to an OOD-exchange slack). Let ψ_s^* be a globally-optimal Stage-1 mixing for prompt set $\mathcal{P}_s$ under any composite reward of the form $F^{(s)}(\lambda) = f_s^{(s)}(\lambda) + \gamma \sum_{j \neq s} f_j^{(s)}(\lambda)$ with $\gamma \geq 0$. Then δ_s^{const} (assigning weight 1 to teacher s at every timestep) is a feasible point, and with the *OOD-exchange term*

$$\Delta_{\text{OOD}} \triangleq \sum_{j \neq s} \left[f_j^{(s)}(\psi_s^*) - f_j^{(s)}(\delta_s^{\text{const}}) \right], \quad (9)$$

the in-domain reward of the optimum satisfies

$$f_s^{(s)}(\psi_s^*) \geq f_s^{(s)}(v^{(s)}) - \gamma \Delta_{\text{OOD}} \geq f_s^{(s)}(v^{(s)}) - \gamma (K-1) R_{\max}, \quad (10)$$

where $R_{\max}$ bounds the per-reward range ($|f_j^{(s)}(\lambda) - f_j^{(s)}(\lambda')| \leq R_{\max}$ on the search space). The slack vanishes as $\gamma \rightarrow 0$, and is only *paid* when the optimum trades in-domain reward for OOD gains ($\Delta_{\text{OOD}} > 0$); if the composite optimum does not improve OOD over δ_s^{const} , then $f_s^{(s)}(\psi_s^*) \geq f_s^{(s)}(v^{(s)})$ with no slack.

Proof. $\delta_s^{\text{const}} \in (\Delta^{K-1})^T$ is in the search space and its induced velocity is exactly $v^{(s)}$, so $f_s^{(s)}(\delta_s^{\text{const}}) = f_s^{(s)}(v^{(s)})$. Optimality of ψ_s^* gives $F^{(s)}(\psi_s^*) \geq F^{(s)}(\delta_s^{\text{const}})$; expanding both sides,

$$f_s^{(s)}(\psi_s^*) + \gamma \sum_{j \neq s} f_j^{(s)}(\psi_s^*) \geq f_s^{(s)}(v^{(s)}) + \gamma \sum_{j \neq s} f_j^{(s)}(\delta_s^{\text{const}}),$$

and rearranging yields $f_s^{(s)}(\psi_s^*) \geq f_s^{(s)}(v^{(s)}) - \gamma \Delta_{\text{OOD}}$ with Δ_{OOD} as in (9). The uniform bound follows from $\Delta_{\text{OOD}} \leq (K-1)R_{\max}$. $\square$

This is the *key advantage* of MoF over OPD-Average: the optimal policy never has to drop below single-teacher in-domain quality.

4.2 Level-2 conditional in-domain strict-improvement

Theorem 2 (Strict in-domain improvement under temporal complementarity). Suppose for some timestep i^* and teacher $k^* \neq s$, $\partial f_s^{(s)} / \partial \lambda_{k^*, i^*} |_{\lambda = \delta_s^{\text{const}}} > 0$ (teacher k^* 's velocity at step i^* contributes positively to R_s). Then there exists a neighborhood of δ_s^{const} inside $(\Delta^{K-1})^T$ such that $f_s^{(s)}(\lambda) > f_s^{(s)}(\delta_s^{\text{const}})$ on a positive-measure subset, and consequently ψ_s^* achieves *strict* in-domain improvement.

Remark 3. This condition—"in some timestep, mixing in another teacher actually helps the in-domain reward"—is what we call **temporal complementarity**. It is an *empirical* property of the teacher ensemble and reward landscape, not a structural property of MoF. Whether it holds depends on whether teachers truly specialize at different denoising stages.

4.3 No general OOD strict-improvement guarantee

The natural follow-up question is: does MoF guarantee that the student *strictly exceeds* every single teacher on every OOD reward?

Theorem 3 (No general OOD guarantee). There exist legitimate MoF instances $(K, T, \{v^{(k)}\}, \{R_j\}, \mathcal{P}_s, \gamma)$ for which, at any optimum ψ_s^* , some OOD reward R_j ($j \neq s$) satisfies

$$f_j^{(s)}(\psi_s^*) < \rho_j^{(s), \text{const}} \triangleq \max_k f_j^{(s)}(\delta_k^{\text{const}}). \quad (11)$$

Thus no framework-only argument (without additional structural assumptions) can prove strict OOD improvement.

Three formal obstructions.

1. **Vertex-maximum block.** If $f_j^{(s)}$ is quasi-concave on $(\Delta^{K-1})^T$ with maximum at a single *constant* vertex $\delta_{k^*}^{\text{const}}$, then no convex combination can exceed $f_j^{(s)}(\delta_{k^*}^{\text{const}}) = \rho_j^{(s), \text{const}}$.
2. **Composite misalignment.** $\psi_s^* = \arg \max F^{(s)}$ generally satisfies $\nabla f_j^{(s)}(\psi_s^*) \neq 0$, so ψ_s^* is *not* a critical point of $f_j^{(s)}$. The composite optimum gives up some f_j to gain $f_s + \sum_{j' \neq j} f_{j'}$.
3. **Reward conflict.** If $\nabla f_s^{(s)} \cdot \nabla f_j^{(s)} < 0$ near δ_s^{const} , the composite optimum lies on a Pareto front and cannot strictly beat both extremes.

Sufficient conditions for OOD strict improvement. Under any one of the following, the negative theorem can be strengthened to a positive guarantee:

- (A) *Temporal complementarity for OOD:* different teachers maximize $\partial f_j^{(s)}/\partial \lambda_{k,i}$ at different timesteps i , AND a Pareto-improving region $\mathcal{Q}_j = \{\lambda : f_s^{(s)}(\lambda) \geq f_s^{(s)}(\delta_s^{\text{const}}) \text{ and } f_j^{(s)}(\lambda) > \rho_j^{(s),\text{const}}\}$ is non-empty;
- (B) *Aligned gradients:* $\nabla f_s^{(s)} \cdot \nabla f_j^{(s)} \geq 0$ near δ_s^{const} (weaker; only guarantees $f_j^{(s)}(\psi^*) \geq f_j^{(s)}(\delta_s^{\text{const}})$, which is not the same as $\rho_j^{(s),\text{const}}$);
- (C) *Pareto-improving region:* $\mathcal{Q}_j \neq \emptyset$ and the composite optimum lands inside it.

A complete proof set, counter-examples (monotonic OOD, vertex-max, distillation-gap), and the deep asymmetry with Level-1 are in `mof_ood_guarantee_analysis.tex`. The asymmetry’s root: Level-1 has a *constructive* fall-back (δ_s^{const}), but OOD has no analogous safe vertex— $\delta_{k^*}^{\text{const}}$ would be the safe vertex for OOD, but it is poor on the in-domain term, so the composite optimization actively avoids it.

4.4 NFT β -cancellation

Theorem 4 (β cancellation). The DiffusionNFT loss (8) satisfies

$$\frac{\partial \mathcal{L}_{\text{NFT}}}{\partial v_{\text{new}}} = A_{\text{max}} \left[r \frac{\partial \mathcal{M}(v^+)}{\partial v^+} - (1-r) \frac{\partial \mathcal{M}(v^-)}{\partial v^-} \right], \quad (12)$$

and this expression *does not depend on* β . Consequently $\partial \mathcal{L}/\partial \psi$ is β -independent (the β in the prefactor of $\mathcal{L}_{\text{NFT}}$ exactly cancels the $\pm\beta$ chain-rule factors from $\partial v^\pm/\partial v_{\text{new}}$). Only the self-normalization weight $w^\pm = \text{sg}(\|x_t - \sigma v^\pm - x_0\|_1/d)$ depends on β , and this enters only as a second-order magnitude rescaling, not direction.

The full derivation is in `nft_beta_analysis.tex`. Practical consequence: β is not a meaningful hyperparameter for MoF; the default $\beta = 1$ should suffice.

4.5 Distillation gap

Stage 2’s MSE (7) is minimized at $v^\theta = v^{\text{MoF}^*}$ on the support of q_θ . Two sources of slack remain:

- **Capacity gap.** v^{MoF^*} may not be in the LoRA-parameterized function class.
- **On-policy distribution shift.** $q_\theta \neq q_{\text{MoF}^*}$; even pointwise-perfect MSE on q_θ does not guarantee equal trajectory-level rewards (ODE integration amplifies small velocity errors).

Theorem 5 (Erosion). Let $\Delta_{\text{stage1}} = f_j^{(s)}(\psi_s^*) - \rho_j^{(s),\text{const}}$ be the Stage-1 OOD margin and $\epsilon_{\text{distill}} = f_j^{(s)}(\psi_s^*) - f_j^{(s)}(\theta^*)$ the distillation gap. If $\epsilon_{\text{distill}} > \Delta_{\text{stage1}}$ then $f_j^{(s)}(\theta^*) < \rho_j^{(s),\text{const}}$, *i.e.* the student’s OOD performance is below the best single teacher despite Stage 1 having achieved strict OOD improvement.

Practical implication: small Stage-1 margins (the typical regime in RL fine-tuning) are vulnerable to erosion. The Stage-2 student capacity (LoRA rank, training budget) must be large enough that $\epsilon_{\text{distill}} \ll \Delta_{\text{stage1}}$.

5 Experimental Program

5.1 Teacher training (preparation, completed)

We trained three Flow-DPPO teachers from a single base flow model, without classifier-free guidance, on three reward objectives:

Teacher	Reward	Selected ckpt	Step	Eval score
teacher_geneval	GenEval	checkpoint-600	1200	0.958
teacher_ocr	OCR	checkpoint-220	440	0.953
teacher_pickscore	PickScore (rescaled $\times 26$)	checkpoint-1500	3000	24.128

These checkpoints serve as the frozen $v^{(k)}$ throughout the MoF experiments. Each teacher’s training curve plateaus near these scores; further fine-tuning yields diminishing returns. The selected checkpoints sit *slightly before* the highest-reward step on each curve—a conservative choice that avoids reward-hacking artifacts that often appear late in DPPO fine-tuning (especially for the OCR objective, where the reward continues climbing past the point of legible-text saturation). Figure 1 shows the train and eval reward trajectories for all three teachers.

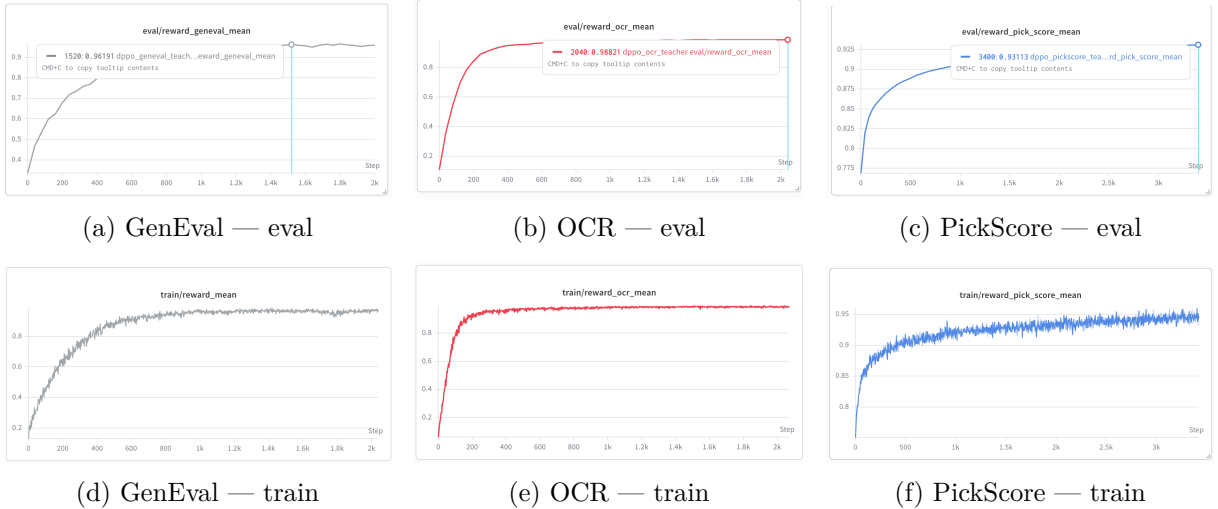


Figure 1: Flow-DPPO teacher training curves (no CFG). **Top row** (a–c): held-out eval reward; **bottom row** (d–f): on-policy training-batch reward. The selected checkpoints used as the frozen $v^{(k)}$ in MoF are GenEval @ step 1200 ($R=0.958$, ckpt-600), OCR @ step 440 ($R=0.953$, ckpt-220), and PickScore @ step 3000 ($R \approx 0.931$ on the rescaled axis, equivalent to 24.128 on the $\times 26$ unrescaled PickScore). These selections are *conservatively earlier* than the peak-reward steps shown in the tooltips; later checkpoints continue to climb in reward but exhibit qualitative degradation (especially OCR overfitting to text-detection heuristics at the cost of overall image quality). All three curves plateau—further single-objective fine-tuning yields diminishing returns, which is precisely the point at which a multi-teacher mixing approach becomes attractive.

5.2 MoF configurations

We ablate two design axes orthogonally:

- **Mixing module:** lut, lut_simple (LUT-SA), time_router, adaln_router, mlp_router.

- **RL algorithm:** DiffusionNFT (default), Flow-GRPO.

The reward configuration follows the standard *per-set composite*

$$A(x_b, s_b) = a_{R_{\text{in}}(s_b)}(x_b) + \gamma \cdot \frac{1}{|J_b|} \sum_{j \in J_b} a_j(x_b), \quad J_b = \{j : j \neq R_{\text{in}}(s_b), a_j(x_b) \neq \text{NaN}\}, \quad (13)$$

with three rewards (GenEval, PickScore, OCR), three sources, and `applicable_sources`: `geneval→[geneval]`
`pick_score→[geneval, pickscore, ocr]` (universal), `ocr→[ocr]`. Default $\gamma = 0.2$.

5.3 Evaluation

We measure both in-domain and OOD performance:

- In-domain: $f_s^{(s)}(\theta^*)$ for each s , vs. the matched single teacher’s eval score above.
- OOD: $f_j^{(s)}(\theta^*)$ for $j \neq s$, vs. each $f_j^{(s)}(v^{(k)})$ baseline (full $K \times S$ cross-table).

6 Empirical findings and ablation plan

6.1 Pathwise vs. PG (verification of Remark 1)

We run the Stage-2 distillation in three modes: `pathwise_only`, `pg_only`, `both`. Hypothesis: `pathwise_only` matches `both` in final reward; `pg_only` fails to converge or has high-variance training. This validates that Stage 2 is a pure regression problem.

6.2 Naive multi-teacher OPD baselines (Step 2)

Two ablations against route-by-source and average-velocity baselines:

- **OPD-route:** target $v^{\text{tgt}}(x, t, p) = v^{(s(p))}(x, t, p)$. Hypothesis: in-domain matches teacher (Level-1 trivial), no OOD gain.
- **OPD-average:** target $v^{\text{tgt}} = \frac{1}{K} \sum_k v^{(k)}$. Hypothesis: in-domain strictly degrades by Jensen, OOD slightly improves.

These provide the lower-bound reference points for MoF.

6.3 MoF main: NFT vs. Flow-GRPO (Step 3)

Run MoF with both RL algorithms, on the LUT and adaLN-router architectures. Hypothesis: NFT converges faster and reaches similar final reward; Flow-GRPO requires more samples but is on-policy. Pick the SOTA per setting.

6.4 Temporal analysis of λ^*

After Stage 1 converges, plot $\lambda_{k,i,s}$ as a function of timestep i for each source s . Hypothesis (informed by analogy with known diffusion phenomena):

- **Early timesteps (high noise):** GenEval-teacher dominates (global layout / object placement).
- **Middle timesteps:** OCR-teacher contributes (text stroke formation).
- **Late timesteps (low noise):** PickScore-teacher dominates (aesthetic refinement).

6.5 PickScore-dominance failure mode in NN routers

Observation: in preliminary runs with neural routers (Time/adaLN/MLP), the learned router consistently routes *all* samples toward `teacher_pickscore`, even on OCR prompts where PickScore-teacher has poor in-domain performance. LUT and LUT-SA do *not* exhibit this failure.

Mechanism (full derivation in `mof_pickscore_dominance_analysis.tex`). Because PickScore is the *only* reward with universal `applicable_sources`, the PickScore signal enters *every* sample’s advantage—once as in-domain (for pickscore samples) and twice via the OOD bonus (for geneval / ocr samples). Total signal coverage on a shared parameter θ :

$$F_{\text{pick}} = \frac{1}{3} \cdot 1.0 + \frac{1}{3} \cdot \gamma + \frac{1}{3} \cdot \gamma = 0.467, \quad (14)$$

$$F_{\text{gen}} = \frac{1}{3} \cdot 1.0 = 0.333, \quad (15)$$

$$F_{\text{ocr}} = \frac{1}{3} \cdot 1.0 = 0.333, \quad (16)$$

so the optimum on a shared-parameter router approximately favors PickScore by $F_{\text{pick}}/F_{\text{ocr}} = 1.4$. The LUT decouples by source, so each column sees only its own 1.0:0.2 ratio and does not develop dominance.

Mitigations (in order of preference):

1. Set $\gamma = 0$ in router runs (lose OOD bonus entirely);
2. Make PickScore domain-specific (`applicable_sources: [pickscore]`);
3. Add a source-conditioning input to the router so it can decouple internally;
4. Re-weight per-source advantage by inverse signal coverage.

Mitigation 1 is the recommended first step.

6.6 γ ablation

Grid over $\gamma \in \{0, 0.05, 0.1, 0.2, 0.5\}$, on adaLN-router and LUT. Plot in-domain and OOD scores. Hypothesis:

- LUT: small γ improvement is monotone within a Pareto-friendly range.
- Router: $\gamma = 0$ recovers OCR; $\gamma > 0$ triggers PickScore-dominance.

This separates the architecture-induced effect from the reward-design effect.

6.7 Initialization ablation

Compare `uniform`, `random`, `teacher_biased` (in-domain teacher initialized to bias $C=2$):

- `teacher_biased`: starts near a Level-1-safe δ_s^{const} vertex; should converge fastest with smallest in-domain dip during early training.
- `uniform`: starts at $1/K$; learning trajectory must overcome the OPD-Average mid-point.
- `random`: control for variance.

6.8 Weight-normalization ablation (softmax / affine / free)

Motivated by Remark 2 (validity needs only $\sum_k \lambda_k = 1$, not convexity) and the softmax mean-subtraction gradient bottleneck (mof_per_set_analysis.tex §13), we ablate the logits→weights mapping on the two neural-router architectures (Time Router and adaLN Router; the final model must rely on network generalization, so LUTs are excluded from this axis). Four arms, implemented via `weight_normalization + weight_sum_penalty`:

- **softmax** (baseline): convex, $\lambda \in \Delta^{K-1}$; gradient $\propto \lambda_j \mathbf{g}^\top (v^{(j)} - \bar{v})$.
- **affine**: hard projection $\lambda = \ell - (\sum_k \ell_k - 1)/K$ onto $\sum \lambda = 1$; lifts only the convexity constraint (CFG-strict analogue). The sum constraint still implies mean-subtraction gradients, so this arm isolates “convexity” from “sum” effects at softmax-scale gradients.
- **free + penalty**: $\lambda = \ell$ raw, with soft anchor $\beta_\Sigma \mathbb{E}_b[(\sum_k \lambda_k(b) - 1)^2]$, $\beta_\Sigma = 0.01$. Gradient $\propto \mathbf{g}^\top v^{(j)}$ (no mean subtraction —largest signal); the penalty suppresses the common-mode drift of $\sum \lambda$ that would rescale the velocity field. LR lowered $\sim 3\times$ (no softmax-Jacobian damping). Recommended default arm.
- **free**: $\lambda = \ell$ with no constraint; measures whether $\sum \lambda$ drift actually occurs and what it costs (monitor `weight_sum_mean/min/max`).

Initialization adapts per mode: routers zero-init the output weight and set the output *bias* to the uniform profile $1/K$ in affine/free modes (an all-zero init would output $\lambda \equiv 0$, i.e. $v_{\text{mix}} = 0$); under softmax the usual zero-init (uniform via logits 0) is kept. AdamW weight decay is disabled for the mixing module in *all* modes —decay anchors weights at 0, which under affine/free shrinks $\sum \lambda$ and rescales the velocity field (the wrong anchor; the sum penalty replaces it).

7 Discussion

7.1 The asymmetry of in-domain vs. OOD

The framework is fundamentally asymmetric: in-domain Level-1 is a *structural* property of the search space (the safe vertex δ_s^{const} is always there, and is composite-friendly), while OOD strict-improvement requires *additional empirical structure* (temporal complementarity, gradient alignment, or non-empty Pareto region). Practitioners should not promise OOD improvement *a priori*; they should measure it.

7.2 When to use which architecture

Briefly:

- **LUT**: best for small T, S (e.g. $K = 3, T = 10, S = 3$); ablation-friendly because each source/timestep slice is independent.
- **LUT-SA**: diagnostic baseline; identifies whether the S -axis genuinely helps.
- **Time Router**: continuous- t flexibility (decouple training and inference T) without prompt conditioning; recommended for inference-step ablations.
- **adaLN/MLP Routers**: prompt-conditioned mixing; provides the most expressive class but inherits the PickScore-dominance pitfall under default γ .

The decision tree: if your reward configuration has a *universal* reward (covering all sources), prefer LUT or set $\gamma = 0$ on the router. If all rewards are domain-specific and you want T -flexibility, the Time Router is a sweet spot.

7.3 Limitations

- Gradient magnitude in NFT can become very small ($\sim 10^{-10}$) under LoRA-shared-base teachers, due to softmax mean-subtraction ($v^{(j)} - \bar{v}$ is a small LoRA residual). Solutions A–D (direct-reward, REINFORCE, residual-space, unnormalized) are catalogued in `mof_per_set_analysis.tex` §13–14.
- Stage 2’s distillation gap is the main practical risk for OOD gains; a multi-round self-distillation refresh of v^{MoF^*} may be needed for small margins.
- All theory is conditional on perfect optimization (global maxima of $F^{(s)}$); in practice, RL with a small number of samples may converge to local optima that violate even the Level-1 fallback.

8 Conclusion

Summary

1. OPD distillation in flow-matching reduces to plain velocity MSE under the (empirically supported, heuristic) assumption that the policy-gradient term contributes only variance (Remark 1). All distillation in this work is therefore a regression problem.
2. Naive multi-teacher OPD has two failure modes (route-by-source: no OOD gain; averaging: in-domain loss). MoF replaces the rigid target with a learnable combination—valid (exactly under shared marginals, approximately for LoRA ensembles; affine suffices, convexity optional) by Prop. 1 and Remark 2, optimizable by RL on a tiny parameter set.
3. Theoretically, MoF gives a *constructive* Level-1 in-domain guarantee (Thm. 1), a conditional Level-2 strict improvement (Thm. 2), and *no* general OOD strict-improvement guarantee (Thm. 3); the asymmetry stems from the absence of a composite-friendly OOD safe vertex.
4. The two-stage pipeline (RL→distill) decouples target optimization from deployment cost. NFT is preferred over Flow-GRPO for faster convergence on the small mixing search; β is irrelevant by Thm. 4.
5. In practice, neural routers can fall into a PickScore-dominance attractor when one reward is universally applicable; LUT and LUT-SA escape this by source-decoupling. Set $\gamma = 0$ or restrict `applicable_sources` to mitigate.

8.1 Open problems

1. Designing a Stage-1 objective with a *constructive* OOD fall-back (e.g. a chance-constrained formulation $\max f_s + \gamma \bar{f}_{\text{OOD}}$ s.t. $f_j \geq \rho_j^{(s), \text{const}}$).
2. Trajectory-level distillation losses that are tighter than pointwise MSE on q_θ , reducing Stage-2 erosion.
3. Identifying necessary and sufficient conditions on the teacher ensemble (LoRA rank, training datasets) under which temporal complementarity *provably* holds.
4. Source-conditioned routers with explicit decoupling priors that combine LUT’s robustness with router’s T -flexibility.

A Pointers to supporting documents

This notes document consolidates the central narrative; technical derivations live in dedicated companion documents:

Topic	File
NFT vs. GRPO loss / gradient comparison	<code>mof_loss_gradient_analysis.tex</code>
β cancellation in NFT	<code>nft_beta_analysis.tex</code>
Per-set λ architecture; gradient vanishing analysis; solutions A/B/C/D	<code>mof_per_set_analysis.tex</code>
Reward / advantage pipeline design	<code>mof_reward_design.tex</code>
Continuous router formulation	<code>mof_continuous_weights.tex</code>
Architecture data-flow (LUT, LUT-SA, Time Router, adaLN, MLP)	<code>mof_router_architectures.tex</code>
PickScore-dominance failure mode analysis	<code>mof_pickscore_dominance_analysis.tex</code>
OOD strict-improvement (negative theorem; sufficient conditions)	<code>mof_ood_guarantee_analysis.tex</code>
End-to-end Stage 1 + Stage 2 pipeline; code map	<code>mof_two_stage_pipeline.tex</code>

References (informal)

When promoting these notes to a paper, fill in full citations for the following works that are inlined as text above:

- **Flow Matching** (Lipman et al., 2023).
- **Flow-GRPO, DiffusionDPO, DiffusionNFT** (recent RL-style diffusion fine-tuning works).
- **OPD** (the on-policy distillation line of work, including the multi-teacher route-by-source variant).
- **Flow-DPPO** (the algorithm used to train the three teachers).